1. The value of $\lim\limits_{x\to 0}\left(\dfrac{x}{\sqrt[8]{1-\sin x}-\sqrt[8]{1+\sin x}}\right)$ is equal to : **[JEE(Main)July2021]**

$\lim\limits_{x\to 0}\left(\dfrac{x}{\sqrt[8]{1-\sin x}-\sqrt[8]{1+\sin x}}\right)$ का मान बराबर है : **[JEE(Main)July2021]**

(A) 4 (B) – 4 (C) – 1 (D) 0

Ans. B

Sol. $\lim\limits_{n\to o}\left(\dfrac{x}{\sqrt[8]{1-\sin x}-\sqrt[8]{1+\sin x}}\right)$

$$\lim_{n\to o}\left(\frac{x}{(1-\sin x)^{\frac{1}{8}}-(1+\sin x)^{\frac{1}{8}}}\right)$$

$$=\lim_{n\to 0}\left(\frac{x}{\left(1-\frac{1}{8}\sin x+\frac{\frac{1}{8}\left(\frac{1}{8}-1\right)}{2!}\sin^2 x-\frac{\frac{1}{8}\left(\frac{1}{8}-1\right)\left(\frac{1}{8}-2\right)}{3!}\sin^3 x.....\right)-\left(1+\frac{1}{8}\sin x+\frac{\frac{1}{8}\left(\frac{1}{8}-1\right)}{2!}\sin^2 x+\frac{\frac{1}{8}\left(\frac{1}{8}-1\right)\left(\frac{1}{8}-2\right)}{3!}\sin^3 x+.....\right)}\right)$$

$$=\lim_{n\to o}\frac{x}{\frac{-2}{8}\sin x-\frac{2\cdot\frac{1}{8}\left(\frac{1}{8}-1\right)\left(\frac{1}{8}-2\right)}{3!}\sin^3 x.....}$$

$$\lim_{n\to o}\frac{x}{\frac{-2}{8}\sin x\left[1+\frac{\left(\frac{1}{8}-1\right)\left(\frac{1}{8}-2\right)}{3!}\sin^2 x+.....\right]}$$

$$=\frac{1}{\frac{-2}{8}\cdot[1+0]}=-4$$

2. If the value of $\lim_{x\to 0}(2-\cos x\sqrt{\cos 2x})^{\left(\frac{x+2}{x^2}\right)}$ is equal e^a, then a is equal to _____.

[JEE(Main)July2021]

यदि $\lim_{x\to 0}(2-\cos x\sqrt{\cos 2x})^{\left(\frac{x+2}{x^2}\right)}$ का मान e^a है, तो a बराबर है _____। **[JEE(Main)July2021]**

Ans. 3

Sol. 1^∞ Form

$$\lim_{x\to 0}(2-\cos x\sqrt{\cos 2x})^{\frac{x+2}{x^2}}=e^{\lim_{x\to 0}\left(\frac{x+2}{x^2}\right)(1-\cos x\sqrt{\cos 2x})}$$

So $a=\lim_{x\to 0}\frac{(x+2)}{x^2}(1-\cos x\sqrt{\cos 2x})$

$$=\lim_{x\to 0}\frac{2}{x^2}\frac{(1-\cos^2 x\cos 2x)}{(1+\cos x\sqrt{\cos 2x})}$$

$$=\lim_{x\to 0}\frac{2(1-\cos^2 x(2\cos^2 x-1))}{2x^2}$$

$$=\lim_{x\to 0}\frac{1+\cos^2 x-2\cos^4 x}{x^2}$$

$$=\lim_{x\to 0}\frac{(1-\cos^2 x)(1+2\cos^2 x)}{x^2}$$

$$=\lim_{x\to 0}\frac{3\sin^2 x}{x^2}=3$$

3. If $\lim\limits_{x\to 0}\dfrac{axe^x-\beta\log_e(1+x)+\gamma x^2e^{-x}}{x\sin^2 x}=10$, $\alpha, \beta, \gamma \in R$ then the value of $\alpha+\beta+\gamma$ is _____.

[JEE(Main)July2021]

यदि $\lim\limits_{x\to 0}\dfrac{axe^x-\beta\log_e(1+x)+\gamma x^2e^{-x}}{x\sin^2 x}=10$, $\alpha, \beta, \gamma \in R$ है, तो $\alpha+\beta+\gamma$ का मान है _____ ।

[JEE(Main)July2021]

Ans. 3

Sol. $\lim\limits_{n\to 0}\dfrac{\alpha x\left(1+x+\dfrac{x^2}{2!}+.....\right)-\beta\left(x+\dfrac{x^2}{2}+\dfrac{x^3}{3}+.....\right)-\gamma x^2\left(1+x+\dfrac{x^2}{2!}-.....\right)}{x^3}$

$\alpha-\beta=0 \qquad \Rightarrow \alpha=\beta$

$\alpha+\dfrac{\beta}{2}+\gamma=0 \qquad \Rightarrow \gamma=\dfrac{-3\beta}{2}$

$\dfrac{\alpha}{2}-\dfrac{\beta}{3}-\gamma=10$

$\beta-6,\ \alpha=6,\ \gamma=-9$

$\alpha+\beta+\gamma=3\ \alpha$

4. $\lim\limits_{x\to 2}\left(\sum\limits_{n=1}^{9}\dfrac{x}{n(n+1)x^2+2(2n+1)x+4}\right)$ is equal to : **[JEE(Main)Aug.2021]**

$\lim\limits_{x\to 2}\left(\sum\limits_{n=1}^{9}\dfrac{x}{n(n+1)x^2+2(2n+1)x+4}\right)$ बराबर है : **[JEE(Main)Aug.2021]**

(A) $\dfrac{9}{44}$ (B) $\dfrac{5}{24}$ (C) $\dfrac{1}{5}$ (D) $\dfrac{7}{36}$

Ans. A

Sol. $\lim\limits_{x\to 2}\sum\limits_{x=1}^{9}\left(\dfrac{x}{x(x+1)x^2+2(2x+1)x+2}\right)$

$\lim\limits_{x\to 2}\sum\limits_{x=1}^{9}\left(\dfrac{x}{(hx+2)x^2+(x+1)x+2}\right)$

$$\lim_{x\to 2}\sum_{x=1}^{9}\left(\frac{1}{hx+2}-\frac{1}{(x+1)x+2}\right)$$

$$\Rightarrow \lim_{x\to 2}\left(\frac{1}{x+2}-\frac{1}{10x+2}\right)=\frac{9}{44}$$

5. If α, β are the distinct roots of $x^2 + bx + c = 0$, then $\lim\limits_{x\to\beta}\frac{e^{2(x^2+bx+c)}-1-2(x^2+bx+c)}{(x-\beta)^2}$ is equal to :

[JEE(Main)Aug.2021]

यदि α, β समीकरण $x^2 + bx + c = 0$ के दो भिन्न मूल हैं, तो $\lim\limits_{x\to\beta}\frac{e^{2(x^2+bx+c)}-1-2(x^2+bx+c)}{(x-\beta)^2}$ बराबर है :

[JEE(Main)Aug.2021]

(A) $2(b^2 + 4c)$ (B) $b^2 - 4c$ (C) $b^2 + 4c$ (D) $2(b^2 - 4c)$

Ans. D

Sol. $$\lim_{x\to\beta}\frac{\left\{1+2(x^2+bx+c)+\frac{\{2(x^2+bx+c)\}^2}{2!}+.....\right\}-1-2(x^2+bx+c)}{(x-\beta)^2}$$

$$\lim_{x\to\beta}\frac{2\left(x^2+bx+c\right)^2}{(x-\beta)^2}=\lim_{x\to\beta}\frac{2(x-\alpha)^2(x-\beta)^2}{(x-\beta)^2}$$

$$\Rightarrow 2(\beta-\alpha)^2=2\left((\alpha+\beta)^2-4\alpha\beta\right)$$

$$=2\left(b^2-4c\right)$$

6. If $\lim\limits_{x\to\infty}\left(\sqrt{x^2-x+1}-ax\right)=b$, then the ordered pair (a, b) is: **[JEE(Main)Aug.2021]**

यदि $\lim\limits_{x\to\infty}\left(\sqrt{x^2-x+1}-ax\right)=b$ है, तो क्रमित युग्म (a, b) है : **[JEE(Main)Aug.2021]**

(A) $\left(-1,\frac{1}{2}\right)$ (B) $\left(-1,-\frac{1}{2}\right)$ (C) $\left(1,\frac{1}{2}\right)$ (D) $\left(1,-\frac{1}{2}\right)$

Ans. D

Sol. $$\lim_{x\to\infty}\sqrt{x^2-x+1}-ax=b$$

$$\lim_{x\to\infty}\frac{x^2-x+1-a^2x^2}{\sqrt{x^2-x+1}+ax}=b$$

$$\lim_{x\to\infty}\frac{x^2(1-a^2)+1-x}{\sqrt{x^2-x+1}+ax}=b$$

If limit exits $a^2=1$

Then $a=\pm1$

$$\lim_{x\to\infty}\frac{1-x}{\sqrt{x^2-x+1}+ax}=b$$

$$\lim_{x\to\infty}\frac{\frac{1}{x}-1}{\sqrt{1-\frac{1}{x}+\frac{1}{x^2}}+a}=b$$

$$\frac{-1}{1+a}=b \qquad (\text{here } a\neq-1)$$

So possible values of b

$$-\frac{1}{2}=b$$

$$\left(1,-\frac{1}{2}\right)$$

7. $\lim_{x\to0}\frac{\sin^2(\pi\cos^4x)}{x^4}$ is equal to : **[JEE(Main)Aug.2021]**

$\lim_{x\to0}\frac{\sin^2(\pi\cos^4x)}{x^4}$ बराबर है : **[JEE(Main)Aug.2021]**

(A) $2\pi^2$ (B) $4\pi^2$ (C) π^2 (D) 4π

Ans. B

Sol. $\lim_{x\to0}\frac{\sin^2(\pi-\pi\cos^4x)}{x^4}\left(\frac{0}{0}\right)$

$$\Rightarrow\lim_{x\to0}\frac{(\pi-\pi\cos^4x)^2}{x^4}$$

$$=\lim_{x\to0}\frac{\pi^2\sin^4x(1+\cos^2x)^2}{x^4}$$

$$=4\pi^2$$

8. If $\alpha = \lim\limits_{x\to\pi/4} \dfrac{\tan^3 x - \tan x}{\cos\left(x+\dfrac{\pi}{4}\right)}$ and $\beta = \lim\limits_{x\to 0}(\cos x)^{\cot x}$ are the roots of the equation, $ax^2 + bx - 4 = 0$, then the ordered pair (a, b) is : **[JEE(Main)Aug.2021]**

यदि समीकरण $ax^2 + bx - 4 = 0$ के मूल $\alpha = \lim\limits_{x\to\pi/4} \dfrac{\tan^3 x - \tan x}{\cos\left(x+\dfrac{\pi}{4}\right)}$ तथा $\beta = \lim\limits_{x\to 0}(\cos x)^{\cot x}$ हैं, तो क्रमित युग्म (a, b) है : **[JEE(Main)Aug.2021]**

(A) (–1, –3) (B) (1, –3) (C) (1, 3) (D) (–1, 3)

Ans. **C**

Sol. $\alpha = \lim\limits_{x\to\frac{\pi}{4}} \dfrac{\tan^3 x - \tan x}{\cos\left(x+\dfrac{\pi}{4}\right)}$

$= \lim\limits_{x\to\frac{\pi}{4}} \dfrac{3\tan^2 x \times \sec^2 x - \sec^2 x}{-\sin\left(x+\dfrac{\pi}{4}\right)}$

$= \dfrac{3\times 1\times 2 - 2}{-1} = -4 = \alpha$

$\beta = \lim\limits_{x\to 0}(\cos x)^{\cot x}$

$= e^{\lim\limits_{x\to 0}\cot x(\cos x - 1)}$

$= e^{\lim\limits_{x\to 0} -\left(\frac{1-\cos x}{\tan x}\right)}$

$= e^{\lim\limits_{x\to 0} -x\left(\frac{1-\cos x}{x^2}\right) = 1 = \beta}$

$-\dfrac{b}{a} = -4 + 1$

$-\dfrac{b}{a} = -3 \Rightarrow \quad b = 3a$

$-\dfrac{4}{a} = -4 \times 1$

$a = 1$

$b = 3$

$(a, b) = (1, 3)$

9. $\lim\limits_{x\to\frac{\pi}{2}}\left(\tan^2 x\left(\left(2\sin^2 x+3\sin x+4\right)^{\frac{1}{2}}-\left(\sin^2 x+6\sin x+2\right)^{\frac{1}{2}}\right)\right)$ is equal to : **[JEE(Main)June2022]**

$\lim\limits_{x\to\frac{\pi}{2}}\left(\tan^2 x\left(\left(2\sin^2 x+3\sin x+4\right)^{\frac{1}{2}}-\left(\sin^2 x+6\sin x+2\right)^{\frac{1}{2}}\right)\right)$ बराबर है : **[JEE(Main)June2022]**

(A) $\frac{1}{12}$ (B) $-\frac{1}{18}$ (C) $-\frac{1}{12}$ (D) $\frac{1}{6}$

Ans. A

Sol. $\lim\limits_{x\to\frac{\pi}{2}}\tan^2 x\left(\sqrt{2\sin^2+3\sin x+4}-\sqrt{\sin^2 x+6\sin x+2}\right)$

$\lim\limits_{x\to\frac{\pi}{2}}\tan^2 x\left(\frac{\left(2\sin^2 x+3\sin x+4\right)-\left(\sin^2 x+6\sin xx+2\right)}{\sqrt{2\sin^2 x+3\sin x+14}+\sqrt{\sin^2 x+6\sin x+2}}\right)$

$\lim\limits_{x\to\frac{\pi}{2}}\frac{\tan^2 x.\left(\sin x-1\right)\left(\sin x-2\right)}{\left(3+3\right)}$

$\lim\limits_{x\to\frac{\pi}{2}}\frac{-1}{6}.\tan^2 x\left(\sin x-1\right)$

put $x=\frac{\pi}{2}-h$

$\lim\limits_{h\to 0}\frac{-1}{6}\cot^2 h\left(\cos h-1\right)$

$\lim\limits_{h\to 0}\frac{1}{6}\frac{\left(1-\cos h\right)}{\tan^2 h}$

$\lim\limits_{h\to 0}\frac{1}{6}\frac{\left(1-\cos h\right)}{h^2}=\frac{1}{6}.\frac{1}{2}=\frac{1}{12}$

10. $\lim\limits_{x\to\frac{1}{\sqrt{2}}}\frac{\sin\left(\cos^{-1}x\right)-x}{1-\tan\left(\cos^{-1}x\right)}$ is equal to : **[JEE(Main)June2022]**

$\lim\limits_{x\to\frac{1}{\sqrt{2}}}\frac{\sin\left(\cos^{-1}x\right)-x}{1-\tan\left(\cos^{-1}x\right)}$ बराबर है : **[JEE(Main)June2022]**

(A) $\sqrt{2}$ (B) $-\sqrt{2}$ (C) $\frac{1}{\sqrt{2}}$ (D) $-\frac{1}{\sqrt{2}}$

Ans. D

Sol. $\lim\limits_{x\to\frac{1}{\sqrt{2}}}\frac{\sin\left(\cos^{-1}x\right)-x}{1-\tan\left(\cos^{-1}x\right)}$

$$\lim_{x\to\frac{1}{\sqrt{2}}}\frac{\sin\left(\sin^{-1}\sqrt{1-x^2}\right)-x}{1-\tan\left(\tan^{-1}\left(\frac{\sqrt{1-x^2}}{x}\right)\right)}$$

$$\lim_{x\to\frac{1}{\sqrt{2}}}\frac{\sqrt{1-x^2}-x}{1-\left(\frac{\sqrt{1-x^2}}{x}\right)}$$

$$\lim_{x\to\frac{1}{\sqrt{2}}}(-x)=-\frac{1}{\sqrt{2}}$$

11. $\lim_{x\to0}\frac{\cos\left(\sin x\right)-\cos x}{x^4}$ is equal to : **[JEE(Main)June2022]**

$\lim_{x\to0}\frac{\cos\left(\sin x\right)-\cos x}{x^4}$ बराबर है : **[JEE(Main)June2022]**

(A) $\frac{1}{3}$ (B) $\frac{1}{4}$ (C) $\frac{1}{6}$ (D) $\frac{1}{12}$

Ans. C

Sol. $\lim_{x\to0}\frac{\cos\left(\sin x\right)-\cos x}{x^4}$

$$\lim_{x\to0}\frac{-2\sin\left(\frac{x+\sin x}{2}\right)\sin\left(\frac{\sin x-x}{2}\right)}{x^4}$$

$$=-2\lim_{x\to0}\frac{\sin\left(\frac{\sin x+x}{2}\right)}{\left(\frac{\sin x+x}{2}\right)}.\frac{\sin\left(\frac{\sin x-x}{2}\right)}{\left(\frac{\sin x+x}{2}\right)}\frac{\left(\sin^2 x-x^2\right)}{4x^2}$$

$$=\frac{-2}{4}\lim_{x\to0}\left(\frac{\sin^2 x-x^2}{x^4}\right);\left(\frac{0}{0}\text{form}\right)\text{L 'Hospitla rule}$$

$$=\frac{-1}{2}\lim_{x\to0}\left(\frac{\sin 2x-2x}{4x^3}\right);\left(\frac{0}{0}\text{form}\right)\text{L 'Hospitla rule}$$

$$=\frac{-1}{2}\lim_{x\to0}\left(\frac{2\cos 2x-2}{12x^2}\right)$$

$$= \frac{-1}{24}\lim_{x\to 0}\left(\frac{2-4\sin^2 x-2}{x^2}\right)$$

$$= \frac{4}{24}\lim_{x\to 0}\left(\frac{\sin^2 x}{x^2}\right)$$

$$= \frac{1}{6}$$

12. Let a be an integer such that $\lim_{x\to 7}\frac{18-[1-x]}{[x-3a]}$ exists, where [t] is greatest integer $\le$ t. Then a is equal to : **[JEE(Main)June2022]**

माना a एक पूर्णांक है जिसके लिए $\lim_{x\to 7}\frac{18-[1-x]}{[x-3a]}$, जहाँ [t] महत्तम पूर्णांक $\le$ t है, का अस्तित्व है, तो a बराबर है : **[JEE(Main)June2022]**

(A) -6 (B) -2 (C) 2 (D) 6

Ans. A

Sol. $\lim_{x\to 7}\frac{18-[1-x]}{[x-3a]}$

$$\lim_{x\to 7}\frac{18-1-[x]}{[x]-3a}$$

$$\lim_{x\to 7}\frac{17-[-x]}{[x]-3a}$$

$$\text{R.H.L.} = \frac{17-(-7)}{6-3a} = \frac{24}{6-3a}$$

$$\text{L.H.L.} = \frac{17-(-8)}{7-3a} = \frac{25}{7-3a}$$

L.H.L. = R.H.L.

$$\frac{24}{6-3a} = \frac{25}{7-3a}$$

$3a = -18$

$a = -6$

13. The value of $\lim_{n\to\infty} 6\tan\left\{\sum_{r=1}^{n}\tan^{-1}\left(\frac{1}{r^2+3r+3}\right)\right.$ is equal to : **[JEE(Main)June2022]**

$\lim_{n\to\infty} 6\tan\left\{\sum_{r=1}^{n}\tan^{-1}\left(\frac{1}{r^2+3r+3}\right)\right.$ का मान बराबर है : **[JEE(Main)June2022]**

(A) 1 (B) 2 (C) 3 (D) 6

Ans. C

Sol. $6\lim\limits_{n\to\infty}\tan\left\{\sum\limits_{r=1}^{n}\tan^{-1}\left(\frac{1}{r^2+3r+3}\right)\right\}$

$=6\lim\limits_{n\to\infty}\tan\left\{\sum\limits_{r=1}^{n}\tan^{-1}\left[\frac{1}{1+\left(r^2+3r+2\right)}\right]\right\}$

$=6\lim\limits_{n\to\infty}\tan\left[\sum\limits_{r=1}^{n}\tan^{-1}\left[\frac{1}{1+(r+2)(r+1)}\right]\right]$

$=6\lim\limits_{n\to\infty}\tan\left[\sum\limits_{r=1}^{n}\tan^{-1}\left[\frac{(r+2)-(r+1)}{1+(r+2)(r+1)}\right]\right]$

$=6\lim\limits_{n\to\infty}\tan\left[\sum\limits_{r=1}^{n}\left(\tan^{-1}(r+2)-\tan^{-1}(r+1)\right)\right]$

$=6\lim\limits_{n\to\infty}\tan\left[\left(\tan^{-1}(n+2)-\tan^{-1}2\right)\right]$

$=6\tan\left[\lim\limits_{n\to\infty}\tan^{-1}(n+2)-\lim\limits_{n\to\infty}\tan^{-1}(2)\right]$

$=6\tan\left[\tan^{-1}(\infty)-\tan^{-1}2\right]=6\tan\left[\frac{\pi}{2}-\tan^{-1}2\right]$

$=\tan\left(\cot^{-2}2\right)$

$=6\tan\left(\tan^{-1}\frac{1}{2}\right)$

$=6\times\frac{1}{2}=3$

14. The value of $\lim\limits_{x\to1}\frac{\left(x^2-1\right)\sin^2(\pi x)}{x^4-2x^3+2x-1}$ is equal to : **[JEE(Main)June2022]**

$\lim\limits_{x\to1}\frac{\left(x^2-1\right)\sin^2(\pi x)}{x^4-2x^3+2x-1}$ का मान बराबर है : **[JEE(Main)June2022]**

(A) $\frac{\pi^2}{6}$ (B) $\frac{\pi^2}{3}$ (C) $\frac{\pi^2}{2}$ (D) π^2

Ans. D

Sol. $\lim\limits_{x\to1}\frac{\left(x^2-1\right)\sin^2(\pi x)}{x^4-2x^3+2x-1}$

$$\lim_{x\to 1}\frac{(x-1)(x+1)\sin^2(\pi x)}{(x-1)^3(x+1)}$$

$$\lim_{x\to 1}\frac{\sin^2(\pi x)}{(x-1)^2}$$

Let $x-1\to t$ $\qquad\qquad \lim_{x\to 0}\frac{\sin^2(\pi(\pi x)}{t^2}$

$x-1\to 0$

$$\Rightarrow \lim_{x\to 0}\frac{\sin^2(t)}{n^2t^2}\pi^2=\pi^2$$

15. Let [t] denote the greatest integer $\le t$ and {t} denote the fractional part of t. The integral value of α for which the left hand limit of the function $f(x)=[1+x]+\frac{\alpha^{2[x]+\{x\}}+[x]-1}{2[x]+\{x\}}$ at $x=0$ is equal to $\alpha-\frac{4}{3}$, is _____.

[JEE(Main)June2022]

माना [t] महत्तम पूर्णांक $\le$ t है, तथा t का भिन्नात्मक भाग {t} है। α का वह पूर्णांक मान, जिसके लिए फलन $f(x)=[1+x]+\frac{\alpha^{2[x]+\{x\}}+[x]-1}{2[x]+\{x\}}$ की $x=0$ पर बाएँ पक्ष की सीमा $\alpha-\frac{4}{3}$ है, है _____

[JEE(Main)June2022]

Ans. 3

Sol. $\lim_{x\to 0}[1+x]+\frac{\alpha^{2[x]+\{x\}}+[x]-1}{2[x]+\{x\}}$

$$\lim_{x\to 0^-}0+\frac{\alpha^{-2+1}-1-1}{-2+1}$$

$$\frac{\alpha^{-1}-2}{-1}=\alpha-\frac{4}{3}$$

$$2-\frac{1}{\alpha}=\alpha-\frac{4}{3}$$

$$\alpha+\frac{1}{\alpha}=\frac{10}{3}$$

$$3\alpha^2-10\alpha+3=0$$

$\alpha=3,\frac{1}{3}$ $\qquad$ Integral value $=3$

16. If $\lim\limits_{x\to 1}\dfrac{\sin\left(3x^2-4x+1\right)-x^2+1}{2x^3-7x^2+ax+b}=-2$, then the value of (a – b) is equal to _____.

[JEE(Main)June2022]

यदि $\lim\limits_{x\to 1}\dfrac{\sin\left(3x^2-4x+1\right)-x^2+1}{2x^3-7x^2+ax+b}=-2$ है, तो (a – b) का मान बराबर है _____

[JEE(Main)June2022]

Ans. 11

Sol. $\lim\limits_{x\to 1}\dfrac{\sin\left(3x^2-4x+1\right)-x^2+1}{2x^3-7x^2+ax+b}=-2$

this will be $\dfrac{0}{0}$ form for finite answer denominator must be zero.

$\because\ 2-7+a+b=0$

$\therefore\ a+b=5$ _____(1)

As this will be $\dfrac{0}{0}$ form use De LHoptal Rule,

$$\lim_{x\to 1}\frac{\cos\left(3x^2-4x+1\right)(6x-4)-2x}{6x^2-14x+a}$$

Again for finite answer denominator must be zero.

$\therefore\ a=8$

from eq. (1)

$b=-3$

required = $a-b=8-(-3)$

$=8+3=11$

17. If $\lim\limits_{n\to\infty}\left(\sqrt{n^2-n-1}+n\alpha+\beta\right)=0$, then $8(\alpha+\beta)$ is equal to : **[JEE(Main)July2022]**

यदि $\lim\limits_{n\to\infty}\left(\sqrt{n^2-n-1}+n\alpha+\beta\right)=0$ है, तो $8(\alpha+\beta)$ बराबर है : **[JEE(Main)July2022]**

(A) 4 (B) –8 (C) –4 (D) 8

Ans. C

Sol. $\lim\limits_{n\to\infty}\left(\sqrt{n^2-n-1}+n\alpha+\beta\right)=0$

$$=\lim_{n\to\infty}\left[\sqrt{1-\frac{1}{n}-\frac{1}{n^2}}+\alpha+\frac{\beta}{n}\right]=0$$

$\therefore\ \alpha=-1$

Now,

$$\lim_{n\to\infty} n\left[\left\{1-\left(\frac{1}{n}+\frac{1}{n^2}\right)\right\}^{1/2}+\frac{\beta}{n}-1\right]=0$$

$$=\lim_{n\to\infty}\frac{\left(1-\frac{1}{2}\left(\frac{1}{n}+\frac{1}{n^2}\right)+.....\right)+\frac{\beta}{n}-1}{\frac{1}{n}}=0$$

$$\Rightarrow \beta-\frac{1}{2}=0$$

$$\therefore \beta=\frac{1}{2}$$

Now, $8(\alpha+\beta)=8\left(-\frac{1}{2}\right)=-4$

18. $\lim\limits_{x\to\frac{\pi}{4}}\frac{8\sqrt{2}-(\cos x+\sin x)^7}{\sqrt{2}-\sqrt{2}\sin 2x}$ is equal to : **[JEE(Main)July2022]**

$\lim\limits_{x\to\frac{\pi}{4}}\frac{8\sqrt{2}-(\cos x+\sin x)^7}{\sqrt{2}-\sqrt{2}\sin 2x}$ बराबर है : **[JEE(Main)July2022]**

(A) 14 (B) 7 (C) $14\sqrt{2}$ (D) $7\sqrt{2}$

Ans. A

Sol. $\lim\limits_{x\to\frac{\pi}{4}}\frac{8\sqrt{2}-\left(\sqrt{2}\right)^7\left(\cos\left(x-\frac{\pi}{4}\right)\right)^7}{\sqrt{2}(1-\sin 2x)}$

$x=\frac{\pi}{4}+h$

$\lim\limits_{h\to 0}\frac{8\sqrt{2}(1-\cos^7 h)}{\sqrt{2}(1-\cos 2h)}$

L–Hospital

$\lim\limits_{h\to 0}8\left(\frac{-7\cos^6 h(-\sinh)}{\sin 2h.2}\right)$

$\lim\limits_{h\to 0}\frac{8\times 7\sinh\cos^6 h}{4\sin\cosh}$

$$= \lim_{h\to 0} \frac{8\times 7}{4}\cos^5 = 14$$

19. Let $f : R \to R$ be a continuous function such that $f(3x) - f(x) = x$. If $f(8) = 7$, then $f(14)$ is equal to : **[JEE(Main)July2022]**

माना $f : R \to R$ एक संतत फलन है जिसके लिए $f(3x) - f(x) = x$ है। यदि $f(8) = 7$ है, तो $f(14)$ बराबर है : **[JEE(Main)July2022]**

(A) 4 (B) 10 (C) 11 (D) 16

Ans. B

Sol. $f(x) - f(x/3) = x/3$

$f(x/3) - f(x/3^2) = x/3^2$

.... on adding

$$f(x) - \lim_{n\to\infty}\left(\frac{x}{3^n}\right) = x\left(\frac{1}{3} + \frac{1}{3^2}....\infty\right)$$

$$f(x) - f(0) = \frac{x}{2}$$

$f(8) = 7$; $f(0) = 3$

$f(x) = x/2 + 3$

$f(14) = 10$

20. Let $\beta = \lim_{x\to 0} \frac{\alpha x - \left(e^{3x} - 1\right)}{\alpha x\left(e^{3x} - 1\right)}$ for some $\alpha \in R$. Then the value of $\alpha + \beta$ is : **[JEE(Main)July2022]**

माना किसी $\alpha \in R$ के लिए $\beta = \lim_{x\to 0} \frac{\alpha x - \left(e^{3x} - 1\right)}{\alpha x\left(e^{3x} - 1\right)}$ है। तो $\alpha + \beta$ का मान है : **[JEE(Main)July2022]**

(A) $\frac{14}{5}$ (B) $\frac{3}{2}$ (C) $\frac{5}{2}$ (D) $\frac{7}{2}$

Ans. C

Sol. $$\beta = \lim_{x\to 0} \frac{\alpha x - \left(e^{3x} - 1\right)}{\alpha x\left(e^{3x} - 1\right)}$$

$$\beta = \lim_{x\to 0} \frac{1 + \alpha x - \left[1 + 3x + \frac{9x^2}{2!} + ...\right]}{(\alpha x)\frac{\left(e^{3x} - 1\right)}{3x}3x}$$

$$\beta = \lim_{x\to 0} \frac{(\alpha x - 3x) - \frac{9x^2}{2!}}{3\alpha x^2}$$

For existence of limit $\alpha - 3 = 0$

$\alpha = 3$

$$\text{Limit } \beta = \frac{-3}{2\alpha}$$

$$\beta = -\frac{1}{2}$$

Now,

$$\alpha + \beta = \frac{5}{2}$$

21. If $\lim\limits_{x\to 0} \frac{\alpha e^x + \beta e^{-x} + \gamma \sin x}{x \sin^2 x} = \frac{2}{3}$, where $\alpha, \gamma \in R$, then which of the following is NOT correct?

[JEE(Main)July2022]

यदि $\lim\limits_{x\to 0} \frac{\alpha e^x + \beta e^{-x} + \gamma \sin x}{x \sin^2 x} = \frac{2}{3}$ है, जहाँ $\alpha, \gamma \in R$ हैं, तो निम्न में से कौन सा सही नहीं है?

[JEE(Main)July2022]

(A) $\alpha^2 + \beta^2 + \gamma^2 = 6$ (B) $\alpha\beta + \beta\gamma + \gamma\alpha + 1 = 0$

(C) $\alpha\beta^2 + \beta\gamma^2 + \gamma\alpha^2 + 3 = 6$ (D) $\alpha^2 - \beta^2 + \gamma^2 = 4$

Ans. C

Sol. $$\lim_{x\to 0} \frac{\alpha\left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!}\right) + \beta\left(1 - x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots\right) + \gamma\left(x - \frac{x^3}{3!} + \ldots\right)}{x^3}$$

constant terms should be zero

$\Rightarrow \alpha + \beta = 0$

coeff of x should be zero

$\Rightarrow \alpha - \beta + \gamma = 0$

coeff of x^2 should be zero

$$\lim_{x\to 0} \frac{x^3\left(\frac{\alpha}{3!} - \frac{\beta}{3!} - \frac{\gamma}{3!}\right) + x^4\left(\frac{\alpha}{3!} - \frac{\beta}{3!} - \frac{\gamma}{3!}\right)}{x^3} = \frac{2}{3}$$

$$\Rightarrow \frac{\alpha}{2} + \frac{\beta}{2} = 0$$

$$\frac{\alpha}{6} - \frac{\beta}{6} - \frac{\gamma}{6} = 2/3$$

$\Rightarrow \alpha = 1, \beta = -1, \gamma = -2$

22. $\lim\limits_{x\to 0}\left(\dfrac{(x+2\cos x)^3+2(x+2\cos x)^2+3\sin(x+2\cos x)}{(x+2)^3+2(x+2)^2+3\sin(x+2)}\right)^{\frac{100}{x}}$ is equal to _____.

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$\lim\limits_{x\to 0}\left(\dfrac{(x+2\cos x)^3+2(x+2\cos x)^2+3\sin(x+2\cos x)}{(x+2)^3+2(x+2)^2+3\sin(x+2)}\right)^{\frac{100}{x}}$ बराबर है_____

[JEE(Main)July2022]

Ans. 1

Sol. $\lim\limits_{x\to 0}\left(\dfrac{(x+2\cos x)^3+2(x+2\cos x)^2+3\sin(x+2\cos x)}{(x+2)^3+2(x+2)^2+3\sin(x+2)}\right)^{\frac{100}{x}}$

Form 1^∞

$= e^{\lim\limits_{x\to 0}\left[\left(\frac{(x+2\cos x)^3+2(x+2\cos x)^2+3\sin(x+2\cos x)}{(x+2)^3+2(x+2)^2+3\sin(x+2)}\right)-1\right]\times\frac{100}{x}}$

$= e^{\lim\limits_{x\to 0}\left[\frac{100}{x}\left(\frac{(x+2\cos x)^3+2(x+2\cos x)^2+3\sin(x+2\cos x)-\left((x+2)^3+2(x+2)^2+3\sin(x+2)\right)}{(x+2)^3+2(x+2)^2+3\sin(x+2)}\right)\right]}$

$= e^{\lim\limits_{x\to 0}\frac{100}{x}\left[\left(\frac{(x+2\cos x)^3+(x+2)^3+2(x+2\cos x)^2-2(x+2)^2+3\sin(x+2\cos x)-3\sin(x+2)}{8+8+3\sin^2}\right)\right]}$

$= e^{\frac{100}{16+3\sin^2}\lim\limits_{x\to 0}\frac{3(x+2\cos x)^2\times(1+2\sin x)-3(x+2)^2-4(x+2\cos x)}{x(1-2\sin x)-4(x+2)+3\cos(x+2\cos x)\times(1-2\sin x)-3\cos(x+2)}}$

$= e^{\frac{100}{16+3\sin 2}\left(\frac{12-3(4)+8\times 1-8+3\cos 2-3\cos 2}{1}\right)}$

Using L'H rule.

$= e^o = 1$